BUSINESS REQUIREMENTS DOCUMENT

Chartly

Guided Charting & Visualization Platform

DOCUMENT VERSION	3.0 — Current Build Baseline
DATE	18 August 2026
STATUS	Approved for MVP Delivery
CLASSIFICATION	Internal — Enterprise Tool
PREPARED BY	Chartly Product & Engineering Team

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1. Executive Summary

Chartly is an internal web-based platform that lets data analysts turn raw CSV or Excel files into interactive, presentation-ready charts in minutes — without leaving the browser and without writing code. Analysts select a dataset, are guided through chart-type, axis, metric, aggregation, sorting, and styling choices, and the system generates an interactive chart using a programmatic rendering engine (Plotly).

The platform is intentionally **AI-free**: no LLM or external provider is involved. All intelligence comes from built-in data profiling, validation rules, and rule-based chart recommendations, which guarantees that organizational data never leaves the environment and that behavior is deterministic and auditable.

Charts are organized into projects, support full version history, can be restored or previewed, exported as PNG/JPG images, and saved as reusable templates. Styling is standardized through a theme-template system (color palettes + fonts), ensuring visual consistency across the team. Every action is recorded in an audit trail.

This document defines the business requirements for the current MVP build and serves as the baseline for further phases.

2. Project Overview

2.1 The Problem

- Analysts spend significant time manually building the same kinds of charts across scattered external tools.
- Outputs are inconsistent: different fonts, colors, and layouts from different people.
- Selections are error-prone: mismatched column types and chart types fail silently or produce misleading visuals.
- There is no central record of how a chart was created or what changed over time.

2.2 The Solution

A single guided workspace that takes an analyst from raw file to reusable chart:

- **Import** — upload CSV or Excel; the system profiles every column (type, uniqueness, missing values, samples).
- **Guide** — dynamic controls adapt to the dataset and chart type; incompatible choices are blocked before generation.
- **Generate** — charts render interactively from a structured specification.
- **Iterate** — edit settings (including colors) and regenerate; every version is preserved.
- **Reuse** — save templates and theme templates (palettes/fonts) and apply them to new datasets with validation.
- **Govern** — role-based permissions and a complete audit trail.

> **Core goal:** minimize the time between raw data and a ready-to-use, consistent, reusable chart — reducing repetitive manual setup while keeping outputs professional and auditable.

3. Business Objectives

1. Reduce the time from raw data to finished chart from hours to minutes.
2. Eliminate reliance on scattered external charting tools.
3. Prevent invalid chart configurations through automatic data profiling and validation.
4. Provide rule-based chart-type recommendations to guide less-experienced users.
5. Ensure visual consistency through unified color palette and font (theme template) management.
6. Build a shared, reusable template library at the team level.
7. Maintain full traceability through version history and an audit trail.
8. Enforce least-privilege access with Viewer / Analyst / Admin roles.

4. Scope of Work

4.1 In Scope (MVP — Current Build)

Area	Deliverables
Data Management	CSV / XLSX / XLS upload (max 50 MB); dataset list with search; rename and delete; row/column metadata; data preview table; per-column profile (type, unique count, missing count, samples).
Guided Chart Builder	10 chart types (bar, line, area, pie, scatter, histogram, box, violin, treemap, funnel); dynamic dimension/metric column lists; aggregation (sum, average, count, min, max); sorting; time granularity (day/week/month/quarter/year) for date dimensions; live configuration summary; rule-based chart recommendations.
Chart Generation & Management	Interactive Plotly rendering; project-based organization; edit-and-regenerate; full version history with diff summaries; preview of any historical version; restore any version.
Styling & Themes	Color palette picker (7 presets + custom colors); font selection; save current palette+font as a theme template; create palettes from scratch (up to 10 colors); apply saved theme templates to any chart; change colors during regeneration.
Export	PNG and JPG image export; Python and JavaScript code export (Plotly) with copy-to-clipboard.
Templates	Save any chart as a reusable template; search by name; apply a template to another dataset with column-compatibility validation; pre-filled builder presets.
Platform Administration	User management (add, delete, role assignment, self-protection); audit trail; platform settings (default chart type); per-user light/dark theme persisted on the account.

Security & Governance	JWT authentication; role-based access control enforced on every action; audit entries for create, regenerate, and restore operations with full specification snapshots.

4.2 Out of Scope (Current Phase)

- Use of an LLM or any external AI provider for chart understanding or generation.
- Sending organizational data to any external service.
- Live database / API data sources — deferred to a later phase.
- Multi-page dashboards and advanced chart types — deferred to a later phase.
- PDF / PPTX / Excel export — current export is image and code only.
- Public template sharing or marketplace — templates are internal to the organization.
- Billing, subscriptions, or multi-tenant SaaS features — the platform is an internal tool.
- Self-service user registration — accounts are provisioned by administrators.

5. Stakeholders, Users & Roles

5.1 Stakeholders

Stakeholder	Interest
Data Analysts	Fast, reliable chart creation and reuse.
Business Stakeholders	Consistent, shareable visuals for decision-making.
Platform Administrators	User provisioning, oversight, and compliance.
IT / Security	Data stays on-premises; no external AI dependencies; auditable actions.

5.2 User Roles and Permissions

Role	Description	Key Permissions
Admin	Owns platform configuration, users, and oversight.	Everything, plus: user management, audit trail, platform settings.
Analyst	Primary creator of datasets, charts, and templates.	Upload data, build/edit/delete charts, regenerate versions, export, save templates and theme templates.
Viewer	Consumer of finished work.	Browse projects, charts, templates; export images and code; no create/edit/delete actions.

> **Enforcement principle:** permissions are checked on the client for usability and enforced server-side on every endpoint. A Viewer cannot perform analyst actions even by calling the API directly.

6. Core User Workflow

1. User signs in; the system applies role-based navigation and permissions.
2. Analyst uploads a CSV or Excel file; the system profiles columns and shows a data preview.
3. Analyst opens the Chart Builder with the dataset pre-selected.
4. The system recommends suitable chart types based on data characteristics.
5. Analyst selects chart type, dimension, metric, aggregation, sorting, granularity, title, palette, and font.
6. The system validates the configuration and shows a summary ("Ready to generate").
7. The chart is generated and displayed interactively on the chart detail page.
8. Analyst reviews; if changes are needed, edits settings — including colors — and regenerates; a new version is recorded.
9. Analyst exports the chart as PNG/JPG or as Python/JavaScript code.

10. Analyst optionally saves the chart as a team template, and/or saves the palette+font as a theme template.
11. Another analyst applies the template to a new dataset; the system validates column compatibility and pre-fills the builder.
12. Administrators monitor all activity in the audit trail and manage users and platform defaults.

7. Functional Requirements

7.1 Authentication & Authorization

ID	Requirement
FR-1.1	The system shall authenticate users via email and password and issue a token-based session (JWT).
FR-1.2	The system shall seed default accounts for each role (Admin, Analyst, Viewer) for evaluation.
FR-1.3	The system shall enforce role-based access on all data, chart, template, and administration endpoints server-side.
FR-1.4	The system shall expire idle sessions and require re-authentication on token failure.

7.2 Data Management

ID	Requirement
FR-2.1	Analysts shall upload CSV, XLSX, or XLS files up to 50 MB.
FR-2.2	The system shall profile each column: detected type (text / number / date), unique count, missing count, and sample values.
FR-2.3	The system shall display a preview table of the first rows after upload and when a dataset is selected.
FR-2.4	Users with delete permission shall remove datasets; the system shall cascade-delete their charts, versions, and audit entries.
FR-2.5	Users shall be able to rename datasets and search the dataset list.

7.3 Guided Chart Builder

ID	Requirement
FR-3.1	The builder shall support bar, line, area, pie, scatter, histogram, box, violin, treemap, and funnel chart types.
FR-3.2	Column lists shall be dynamic: dimensions include all columns; metrics include numeric columns only (except count aggregation).
FR-3.3	The builder shall support aggregation functions: sum, average, count, min, max; aggregation shall be hidden for raw-data charts (histogram, box, violin).
FR-3.4	The builder shall support sorting (none, ascending, descending) and time granularity (day–year) when the dimension is a date and the chart type is bar, line, or area.
FR-3.5	The system shall display context-sensitive hints per chart type and a live configuration summary before generation.

FR-3.6	The system shall provide rule-based chart-type recommendations per dataset with a human-readable reason for each.
FR-3.7	Selected theme templates and palettes shall pre-fill the style controls when entering the builder.

7.4 Chart Generation, Regeneration & Versions

ID	Requirement
FR-4.1	The system shall generate charts from a structured specification (chart type, dimension, metric, aggregation, sorting, granularity, title, colors, font).
FR-4.2	Charts shall render interactively (hover, zoom, pan, download menu) using Plotly.
FR-4.3	Analysts shall edit any specification field — including colors — and regenerate; each regeneration shall create a new version with a timestamp.
FR-4.4	The version history shall show which fields changed between consecutive versions.
FR-4.5	Any historical version shall be previewable in a modal and restorable; restoring shall create a new version entry so history is never lost.
FR-4.6	Charts shall be exportable as PNG and JPG images at high resolution.
FR-4.7	The system shall generate exportable Python and JavaScript code reproducing the chart, with copy-to-clipboard.

7.5 Styling & Theme Templates

ID	Requirement
FR-5.1	The builder shall offer seven preset color palettes plus a fully custom palette with four pickers.
FR-5.2	Users shall save the current palette and font as a named theme template.
FR-5.3	Users shall create a new palette from scratch (up to 10 colors) via a dedicated modal and save it directly as a theme template.
FR-5.4	Users shall apply a saved theme template to the current chart in one click; the palette and font shall be reflected in the generated chart.
FR-5.5	Theme templates shall be listed with a visual palette preview, searchable, deletable, and reusable from the Templates page.
FR-5.6	Colors shall be validated as hex values; a maximum of 10 colors per palette shall be enforced.

7.6 Template Library

ID	Requirement
FR-6.1	Analysts shall save any chart as a reusable template with a name.

FR-6.2	Templates shall be searchable by name.
FR-6.3	Applying a template to a dataset shall validate that the required columns exist and are type-compatible, then pre-fill the builder.
FR-6.4	Template application shall carry over style settings (colors, font) into the builder preset.

7.7 Administration

ID	Requirement
FR-7.1	Admins shall create, update (name/role/password), and delete user accounts.
FR-7.2	The system shall prevent an admin from demoting or deleting their own account, and shall prevent deletion of the last admin.
FR-7.3	Admins shall set the platform's default chart type used when the builder opens.
FR-7.4	The audit trail shall record user, action (create / regenerate / restore), chart, timestamp, and full specification for every event, viewable by admins.
FR-7.5	Each user's light/dark theme preference shall be persisted on their account and applied on every device.

7.8 Dashboard

ID	Requirement
FR-8.1	The dashboard shall show KPIs: datasets, charts created, templates, and datasets added this week.
FR-8.2	The dashboard shall provide quick actions (Upload Data, Create Chart, Templates) and a recent activity feed sourced from the audit trail.

8. Non-Functional Requirements

Category	Requirement
Security	Passwords hashed with salted SHA-256; tokens signed and expiring; no secrets in client code; CORS restricted to configured origins.
Data Privacy	All data processing is in-process — no external AI providers, no data leaves the deployment environment.
Access Control	Role checks enforced server-side on every protected endpoint; client-side UI adapts to the role.
Auditability	Every create/regenerate/restore action persists a specification snapshot with user and timestamp.
Performance	Uploads up to 50 MB; interactive chart rendering; version lists render without pagination at expected team scale.
Compatibility	Current versions of Chromium-based browsers (Chrome, Edge) and Firefox; responsive layout from 1024 px up to 1920 px without horizontal page overflow.
Reliability	Core MVP functions operate without any external AI or cloud charting dependency; charts render from local data only.
Maintainability	Backend organized by service modules (profiling, validation, generation) and versioned API; frontend modularized into reusable views and components.

9. UI / UX Requirements

- **Layout:** fixed-width sidebar navigation; content area fills remaining viewport; no page-level horizontal scrolling at desktop sizes; wide preview tables scroll inside their own container only.
- **Consistency:** single design system (orange primary, indigo secondary), shared cards, buttons, badges, forms, toasts, and modals across all pages.
- **Guidance:** builder shows per-chart-type hints, a live configuration summary, and rule-based recommendations before generation.
- **Feedback:** toast notifications for every success/failure; loading states and progress bars during uploads and generation.
- **Theme:** full light/dark mode, applied instantly and remembered per user.
- **Accessibility baseline:** semantic buttons/links, visible focus states, contrast within the design system, keyboard-operable modals.

10. Risks and Mitigations

Risk	Impact	Mitigation
Unclean or malformed files break profiling (mixed types, missing headers, dates as text).	Medium	Robust type detection with fallback classification; clear error messages; count aggregation works on non-numeric columns.
Large datasets degrade upload or rendering.	Medium	50 MB cap; preview limited to first rows; aggregation happens server-side on the full frame.
Wide tables or long column names cause layout overflow.	Low	Responsive grid constraints; contained table scrolling; wrapping schema chips.
Templates reference columns missing in a target dataset.	Medium	Server-side column validation before applying a template; clear error feedback.
Data volume / concurrency targets not yet measured.	Low	PoC with realistic team data before scaling decisions.
Version history grows without bound.	Low	Retention policy to be evaluated in Phase 2; restore preserves history.

11. Implementation Phases

Phase 1 — MVP DELIVERED

- CSV/Excel upload with profiling and preview.
- Guided builder with 10 chart types, aggregation, sorting, granularity, recommendations.
- Interactive generation, edit-and-regenerate, version history with preview/restore.
- PNG/JPG and code export.
- Template library and theme templates (palettes + fonts) with validation.
- Viewer / Analyst / Admin roles, audit trail, platform settings, per-user theming.
- Responsive layout fixes at 1024–1920 px.

Phase 2

- Self-service profile: display name and password change from Settings.
- Live data sources (database/API connectors).
- Dashboards combining multiple charts; advanced chart types.
- Audit and version retention policies; PDF/PPTX export.
- Extended recommendation engine and template mapping improvements.

Phase 3

- Performance optimization and horizontal scaling.
- Team-level statistics and analytics on usage.

- Optional natural-language guidance — only if business need and privacy requirements allow, never sending data externally.

12. Success Criteria and KPIs

Criterion	Target
Time from raw data to finished chart	Reduce from hours to minutes (≤ 10 minutes end-to-end).
First-attempt generation success rate	≥ 90% of generated charts are accepted without configuration changes.
Validation error rate	No charts generated from incompatible column/chart-type selections.
Template adoption	≥ 50% of analysts reuse at least one saved template or theme template monthly.
External tool usage	Charts for internal reporting produced inside Chartly.
Audit completeness	100% of create/regenerate/restore actions recorded with specification snapshots.
Availability	Core workflow available during business hours with zero data-loss incidents.

13. Assumptions, Dependencies & Constraints

13.1 Assumptions

- Users have modern browsers (Chrome, Edge, or Firefox) and no external software is required.
- Analysts receive structured or semi-structured tabular files; arbitrary free-form documents are out of scope.
- The platform is deployed inside the organization; internet connectivity is not required for chart generation.

13.2 Dependencies

- Python 3.12+ runtime with FastAPI, SQLAlchemy, pandas, and Plotly libraries.
- SQLite database for the MVP deployment (upgradable to a production DB in later phases).
- Plotly CDN loaded by the browser for interactive rendering.

13.3 Constraints

- No external AI/LLM services may receive organizational data.
- Maximum upload size: 50 MB; supported formats: CSV, XLSX, XLS.
- Image export formats limited to PNG and JPG in this phase.
- The sidebar brand name is fixed by the product team and not user-editable in Settings.

14. Document Control & Approval

14.1 Revision History

Version	Date	Author	Summary of Changes
1.0	Draft	Product Team	Initial concept — guided chart generation.
2.0	10 Aug 2026	Product Team	No-LLM architecture; validation and recommendation rules; template library; roles.
3.0	18 Aug 2026	Product & Engineering	Current build baseline: 10 chart types, theme templates & palette management, version restore/preview, code export, responsive layout, platform settings.

14.2 Sign-off

Role	Name	Date
Product Owner	_____	_____
Business Stakeholder	_____	_____
Engineering Lead	_____	_____